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by Hasso-Plattner-Institut

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für Forschung, Technologie
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# Effizientes Abfragen von Informationen

## Lokales Retrieval Augmented-Generation System

Hanno Müller

Design IT.
Create Knowledge.

www.hpi.de

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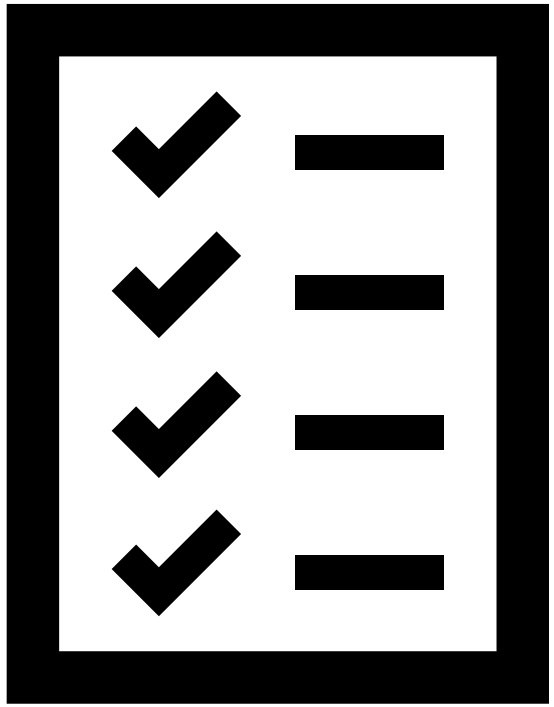
# Anwesenheitsliste



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# LEARNING GOALS

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# Motivation

- LLMs warden (größtenteils) mit öffentlichen Daten trainiert. Daher haben LLMs kein Wissen über **private Daten**, z.b.
  - **interne Dolumente** über Betriebsabläufe
  - **zahlungspflichtige Dokumente** wie wissenschaftliche Artikel
- LLMs vermischen Daten. Wenn keine **relevante Information** vorliegt, um die Fragen zu beantworten, warden ähnliche Informationen verwendet, z.B.
  - Rechtstexte zu Parkverstößen warden benutzt um Fragen zu Halteverstößen zu beantworten
  - Populärwissenschaftliche Texte warden verwendet um wissenschaftliche Fragen zu beantworten
- RAG ermöglicht die Verwendung **vordefinierter Documente** zur Fragenbeantwortung!

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# Lernziele

- Verständis für die Grundlangen von RAG entwickeln
  - Embeddings
  - Vector Databases
  - Information Retrieval
  - Prompt Engineering
- Anforderungen an (eigene) RAG-Systeme verstehen
  - z.b., Datenaufbereitung
  - z.b., Vektor-Datenbanken erstellen
- Deployes eines lokalen RAG-Systems

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# Agenda

1. Allgemeine Einleitung (30 minutes)
2. Demonstration von RAG (30 minutes)
3. Interaktives Notebook (30 minutes)
4. Deployen von RAG + Pause (30 minutes)
5. RAG indivudell konfigurieren (30 minutes)
6. Diskussion (30 minutes)

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# FUNDAMENTALS

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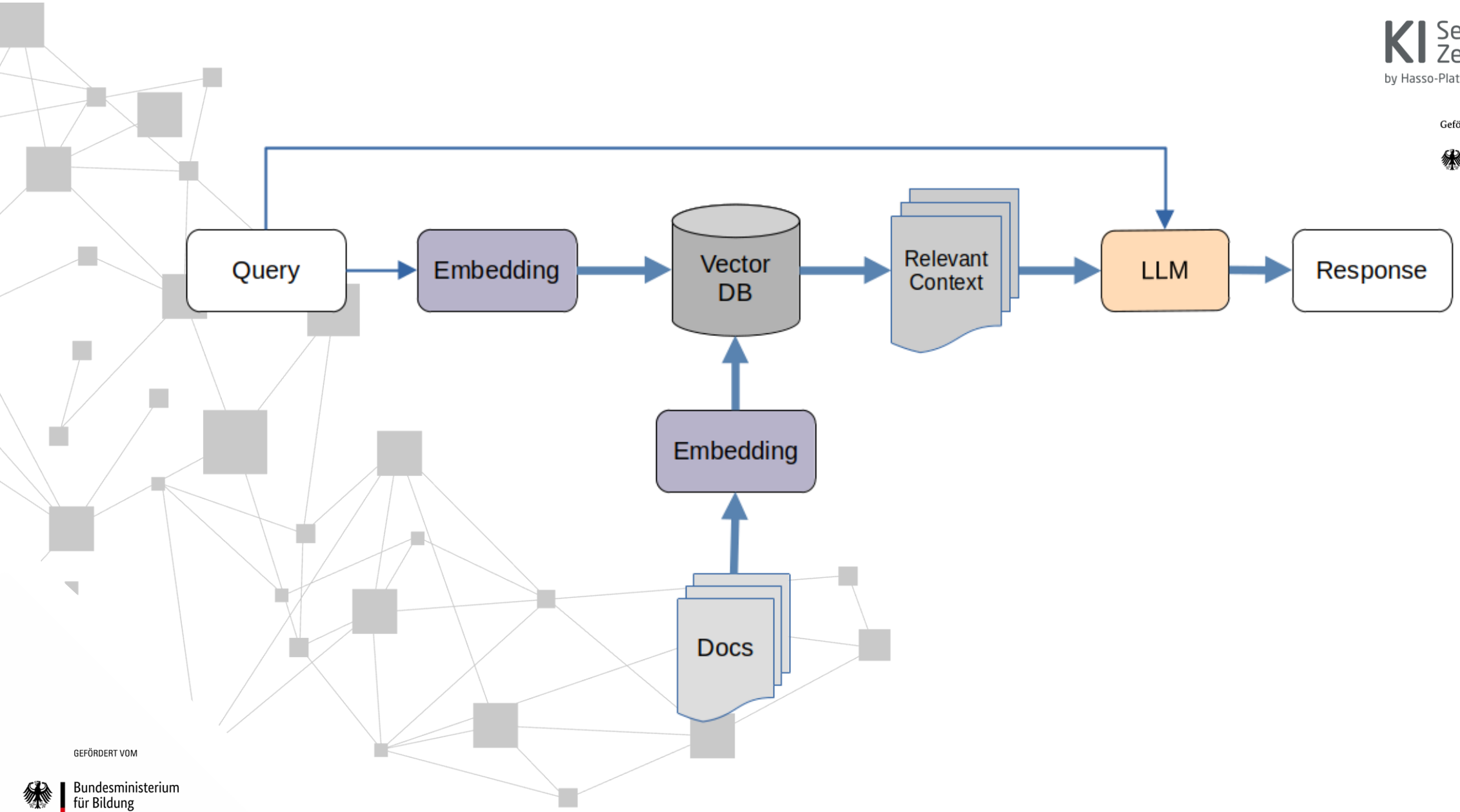


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GEFÖRDERT VOM
Bundesministerium für Bildung und Forschung

# Local RAG Tool

RAG Query | Semantic Retrieval | RAG Chat | Chat | Rag Tool Settings

Query an LLM about information from your documents.

Query

Settings

Submit

Answer

Export

# Hands-On

https://github.com/aihpi/workshop-rag/blob/main/learning-materials/01_workflow.ipynb

https://github.com/aihpi/

→ repositories

→ workshop-rag

→ learning-materials

→ 01_workflow.ipynb

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# EXERCISE

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# Übung

1. Lade deine eigenen Dokumente in die Vektor-Datenbank

2. Passe den Prompt so an, dass eine Stichwortliste als Antwort ausgegeben wird, z.B.

   Frage:
   Wer ist Watson?

   Antwort:
   - Eine fiktionale Figur in den Sherlock Holmes Geschichten
   - Ein Freund von Holmes
   - Erschien zum ersten Mal in “A Study in Scarlet” (1887)

3. Optimiere die Parameter Relevance Threshold, Top K, und Temperatur für verbesserte Antworten. Was machen diese Parameter?

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# DISCUSSION

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# Vier AI Service-Zentren in Deutschland

KI Service Zentrum by Hasso-Plattner-Institut

https://hpi.de/ki-servicezentrum/

## Überblick

KI-Servicezentren

**Ziel:**
**Reduktion von Barrieren bei der Implementierung von KI-Anwendungen in Gesellschaft und Wirtschaft.**

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# KI-Servicezentrum Berlin Brandenburg

KI Service Zentrum by Hasso-Plattner-Institut HPI

**Struktur**

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# Forschung

- **KI-Methodenforschung**
- **KI-Betriebsforschung**

**PubMedCLIP: How Much Does CLIP Benefit Visual Question Answering in the Medical Domain?**


Sedigheh Eslami, Christoph Meinel, Gerard de Melo

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Abstract

Contrastive Language–Image Pre-training (CLIP) has shown remarkable success in learning with cross-modal supervision from extensive amounts of image–text pairs collected online. Thus far, the effectiveness of CLIP has been investigated primarily in general-domain multimodal problems. In this work, we evaluate the effectiveness of CLIP for the task of Medical Visual Question Answering (MedVQA). We present PubMedCLIP, a fine-tuned version of CLIP for the medical domain based on PubMed articles. Our experiments conducted on two MedVQA benchmark datasets illustrate that PubMedCLIP achieves superior results improving the overall accuracy up to 3% in comparison to the state-of-the-art Model-Agnostic Meta-Learning (MAML) networks pre-trained only on visual data. The PubMedCLIP model with different back-ends, the source code for pre-training them and reproducing our MedVQA pipeline is publicly available at https://github.com/sarahESL/PubMedCLIP.


1 Introduction

Medical visual question answering (MedVQA) seeks answers to natural language questions about a given medical image. The development of MedVQA has considerable potential to benefit healthcare systems, as it may aid clinicians in interpreting medical images and obtaining more accurate diagnoses by consulting a second opinion. Thus, it has become a very active area of research, with competitive benchmarks and yearly competitions (Abacha et al., 2021). Yet, visual question answering in the medical domain in particular remains non-trivial as we suffer from a general lack of large balanced training data, in part due to privacy concerns. To solve the multimodal task of MedVQA, a system must understand both medical images and textual questions and infer the associations between them sufficiently well to produce a correct answer (An

*Findings of the Association for Computati*
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**Exploring Paracrawl for Document-level Neural Machine Translation**


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Abstract

Document-level neural machine translation (NMT) has outperformed sentence-level NMT on a number of datasets. However, document-level NMT is still not widely adopted in real-world translation systems mainly due to the lack of large-scale general-domain training data for document-level NMT. We examine the effectiveness of using Paracrawl for learning document-level translation. Paracrawl is a large-scale parallel corpus crawled from the Internet and contains data from various domains. The official Paracrawl corpus was released as parallel sentences (extracted from parallel webpages) and therefore previous works only used Paracrawl for learning sentence-level translation. In this work, we extract parallel paragraphs from Paracrawl parallel webpages using automatic sentence alignments and we use the extracted parallel paragraphs as parallel documents for training document-level translation models. We show that document-level NMT models trained with only parallel paragraphs from Paracrawl can be used to translate real documents from TED, News and Europarl, outperforming sentence-level NMT models. We also perform a targeted pronoun evaluation and show that document-level models trained with Paracrawl data can help context-aware pronoun translation. We release our data and code here[1].


1 Introduction

The Transformer translation model (Vaswani et al., 2017), which performs sentence-level translation based on attention networks, has achieved great success and significantly improved the state-of-the-art in machine translation. Compared to sentence-level translation, document-level translation (Xu et al., 2021; Bao et al., 2021; Jauregi Unanue et al., 2020; Ma et al., 2020; Maruf et al., 2019; Tu et al., 2018; Maruf and Haffari, 2018) performs translation at document-level and can potentially fur-

[1]https://github.com/Yusser96/Exploring-Paracrawl-for-Document-level-Neural-Machine-Translation

*Proceedings of the 17th Conference of the European Chapter of*
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**Efficient Parallelization Layouts for Large-Scale Distributed Model Training**


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Abstract

Efficiently training large language models requires parallelizing across hundreds of hardware accelerators and invoking various compute and memory optimizations. When combined, many of these strategies have complex interactions regarding the final training efficiency. Prior work tackling this problem did not have access to the latest set of optimizations, such as FlashAttention or sequence parallelism. In this work, we conduct a comprehensive ablation study of possible training configurations for large language models. We distill this large study into several key recommendations for the most efficient training. For instance, we find that using a micro-batch size of 1 usually enables the most efficient training layouts. Larger micro-batch sizes necessitate activation checkpointing or higher degrees of model parallelism and also lead to larger pipeline bubbles. Our most efficient configurations enable us to achieve state-of-the-art training efficiency results over a range of model sizes, most notably a Model FLOPs utilization of 70.5% when training a Llama 13B model.


1 Introduction

The number of parameters and computational resources spent on training deep neural networks is growing rapidly [1, 3, 14]. The largest models consisting of hundreds of billions of parameters do not even fit onto a single hardware accelerator. Thus, training these models requires various ways of reducing the memory requirements, such as ZeRO [16], activation checkpointing [2], and 3D-parallel (data, tensor, and pipeline parallel) training [13]. 3D parallelism, in particular, has been demonstrated to be effective for the training of Transformer-based large language models (LLMs) with hundreds of billions of parameters [13].

However, training these models efficiently with 3D parallelism requires significant domain expertise and extensive manual effort to determine the ideal configurations. These configurations not only need to combine data, model, and pipeline parallelism most efficiently, but also consider complex interactions with other memory and compute optimizations. FlashAttention [5] in particular has had a notable impact since its release, enabling us to train models at previously impossible degrees of training efficiency. In light of these developments, we conduct a systematic study via a large-scale training efficiency sweep of these interactions. We consider up to 256 GPUs and Llama models with up to 65 billion parameters.

Workshop on Advancing Neural Network Training at 37th Conference on Neural Information Processing Systems (WANT@NeurIPS 2023).


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Zugangsanfrage
aisc.hpi.de

# Infrastruktur

- **Kostenloser Zugang**
- Kein Produktionsbetrieb
  - Daten sollten **anonymisier** oder **synthetisierd** werden
  - Kein **hosting** von Produkten
- **Berichterstattung** & **Veröffentlichung** durch Nutzer:innen
- **Alte Rechte** verbleiben bei Nutzer:innen
- **Neue Rechte** verbleiben bei Nutzer:innen
  - Gewährung von Nutzungsrechten für Forschung und Lehre

**Training**
- 64 NVIDIA H100 GPU

**Inferenz**
- 40 NVIDIA A30 GPU

**ARM Server**
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

**GPU Server**
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

**Edge**
- ARMv8 CPU
- NVIDIA Jetson AGX Module

**Neuromorph**
- 288 SpiNNaker2 Chips

**Storage**
- 1.5 PB NVMe

**Network**
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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Newsletter

# Bildung

## Talks

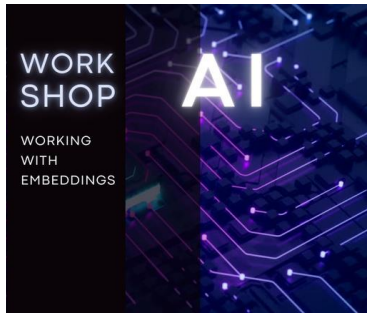
tele-task.de/series/1463

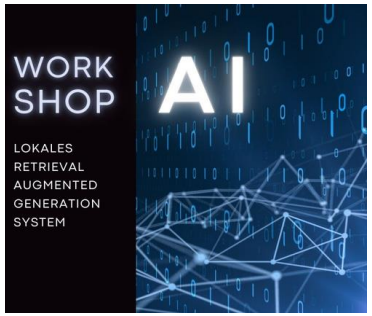
- Gastvorträge über Forschung und Innovation

## Work shops

aimaker.community

- Praktische Themen
- Beispielthemen: Speech2summary, Docker für ML, Semantische Suche

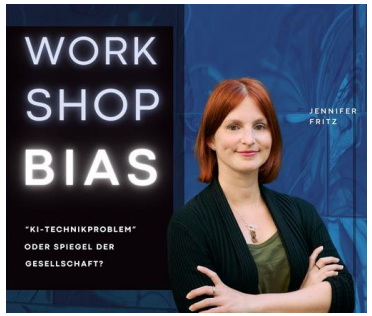


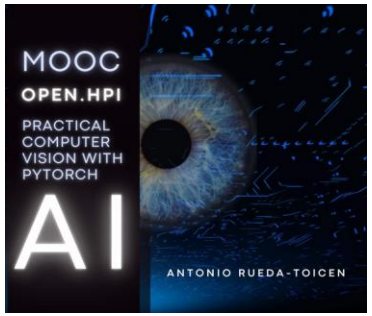



## MOOCs

open.hpi.de/channels/ai-service-center

- ChatGPT: Was bedeutet generative KI für die Gesellschaft?
- Profitable AI
- KI Biases verstehen und vermeiden



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Sprechstunde buchen

# Consulting

## AI consultation hours

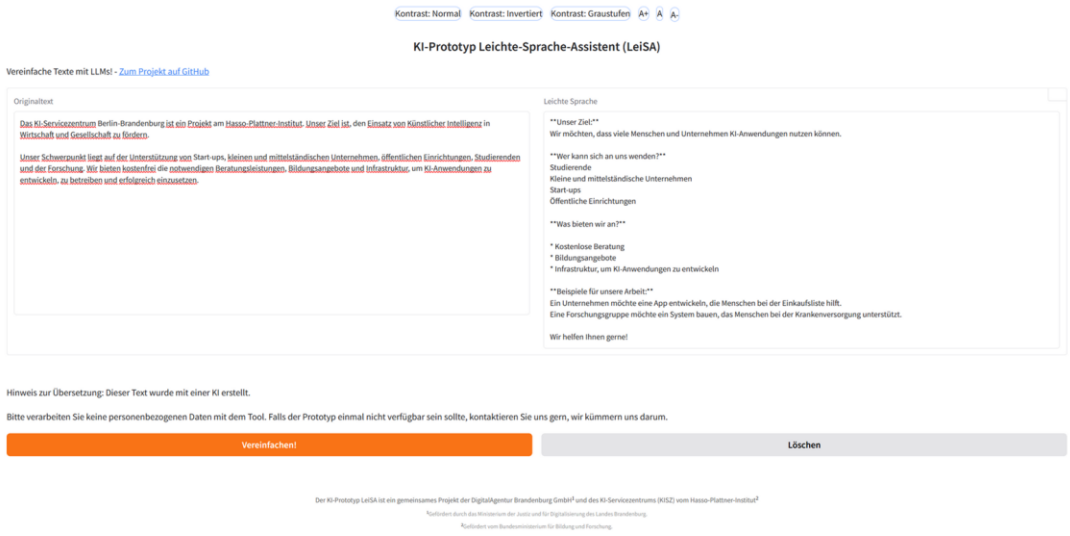
- Answering questions:
  - about AI infrastructure
  - about AI models & frameworks
  - about AI use cases

## AI pilot projects

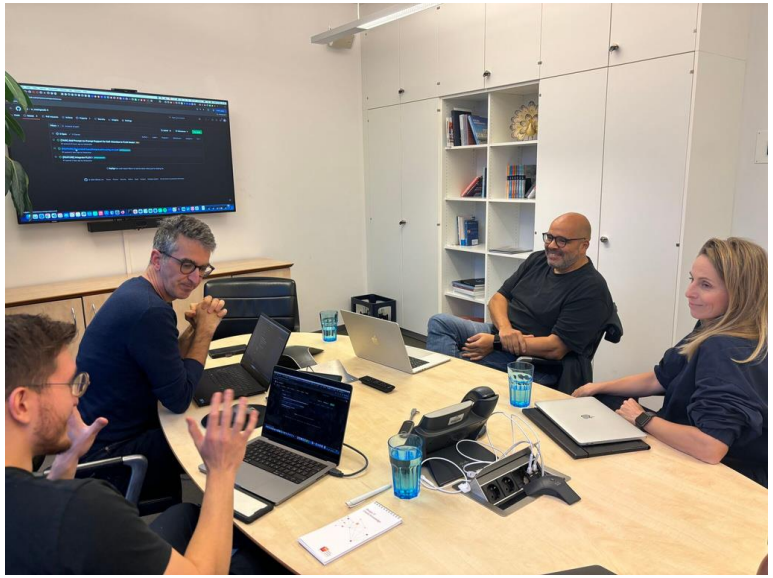
- Co-development of a prototype
- Application every three months
- Selection criteria, e.g. AI maturity, public benefit
- Publication of results

## Cooperation

- Jointly organized network meetings




github.com/aihpi/leichte-sprache



**Previous AI pilot projects**

- Generation of math problems
- Generation of easy language
- Generation of upcycling suggestions
- Reduction of food waste
- Dating by handwriting

Jetzt bewerben!

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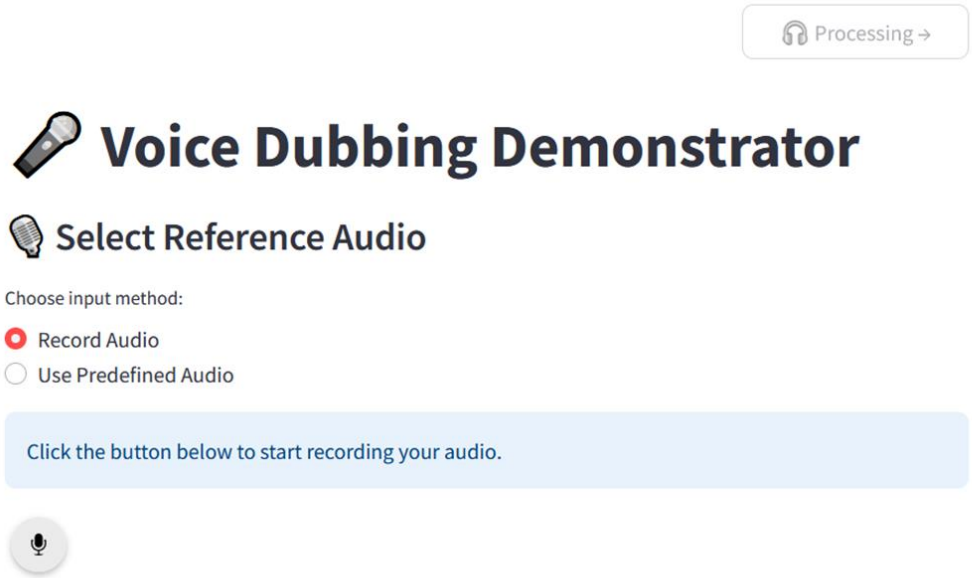
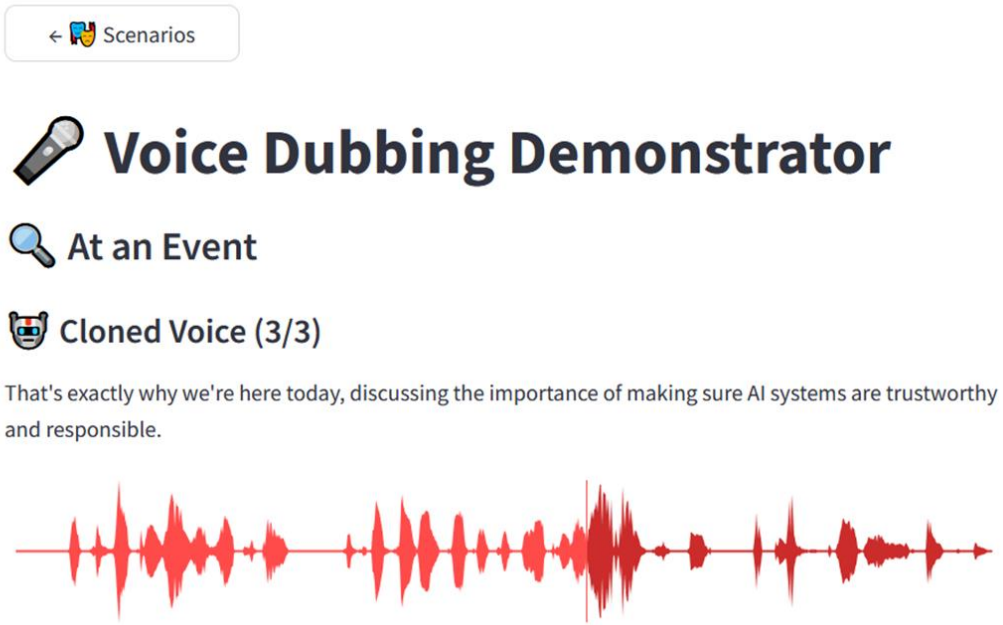
# Verarbeitung gesprochener Sprache

https://tts.aisc.hpi.de/
user: event
pw: aisc@hpi2022!

## Prototypen

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# Deine Meinung ist wichtig!

QR code zum Feedback Formular

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